

WIRELESS LINK BUDGET ANALYSIS REPORT

Course: 3171608 - Wireless Communication (WC) | Micro-Project

Scenario: GEO Satellite Ku-band TV / Data Downlink | Generated: 2026-10-05 23:29:38 | Model: Free Space Path Loss (FSPL)

LINK STATUS: ✓ PASS — EXCELLENT LINK (Margin >= 10 dB)
Received Power: -90.05 dBm | Sensitivity: -115.00 dBm | Fade / Link Margin: +24.95 dB

Executive Performance Metrics

EIRP	86.50 dBm (446683.592 W)	Received Power (Prx)	-90.05 dBm (0.99 pW)
Free Space Path Loss	205.15 dB	Model Path Loss	205.15 dB
Total Environmental Loss	9.90 dB	Total Attenuation	217.55 dB
Thermal Noise Floor	-99.78 dBm	Signal-to-Noise Ratio (SNR)	9.73 dB
Max Achievable Distance	50000.000 km	Shannon Capacity	121.56 Mbps
Max Allowable Path Loss	230.10 dB	Carrier Wavelength (λ)	0.0250 m (2.50 cm)

System Input Parameters

Subsystem	Parameter Name	Value	Units / Notes
Transmitter (Tx)	Tx Output Power (Ptx)	50.00	dBm
	Tx Antenna Gain (Gtx)	38.00	dBi
	Tx Cable & Connector Losses	1.50	dB
	Tx Antenna Height (hb)	30.0	meters
Channel / Path	Carrier Frequency (f)	12000.00	MHz
	Link Distance (d)	36000.000	km
	Propagation Model	Free Space Path Loss (FSPL)	Empirical/Analytical
	Path Loss Exponent (n)	3.00	Dimensionless
	Rain Attenuation	3.00	dB
	Atmospheric & Gas Loss	0.60	dB
	Obstacle / Penetration Loss	0.00	dB
	Fade Margin Allocation	6.00	dB
Receiver (Rx)	Rx Antenna Gain (Grx)	39.50	dBi
	Rx Cable & Filter Loss	1.00	dB
	Rx Sensitivity Threshold	-115.00	dBm
	Front-End Noise Figure (NF)	1.50	dB

	Channel Bandwidth (B)	36.00	MHz
	Noise Temperature (T)	150.0	Kelvin (K)

Link Budget Stage-by-Stage Power Accounting

Stage Description	Incremental Gain/Loss (dB)	Cumulative Power Level (dBm)
Tx Power	+50.00 dB	50.00 dBm
After Tx Loss	-1.50 dB	48.50 dBm
EIRP (Tx Antenna)	+38.00 dB	86.50 dBm
After Path Loss	-205.15 dB	-118.65 dBm
After Env Losses	-9.90 dB	-128.55 dBm
Rx Antenna Gain	+39.50 dB	-89.05 dBm
Rx Power (Input)	-1.00 dB	-90.05 dBm

- Core Mathematical Formulas Applied:
- **Friis Received Power:** $Prx\text{ (dBm)} = Ptx + Gtx - Ltx - PL + Grx - Lrx - Lenv$
 - **Free Space Path Loss:** $FSPL\text{ (dB)} = 32.44 + 20 \cdot \log_{10}(d_km) + 20 \cdot \log_{10}(f_MHz)$
 - **Thermal Noise Floor:** $N\text{ (dBm)} = -174\text{ dBm/Hz} + 10 \cdot \log_{10}(B_Hz) + NF\text{ (dB)}$
 - **Link Margin:** $Margin\text{ (dB)} = Prx\text{ (dBm)} - Prx,sens\text{ (dBm)}$
 - **Shannon Capacity:** $C = B \cdot \log_2(1 + SNR_linear)\text{ (bps)}$